

Prompt Engineering and Large Language Models

Quiz 2

Weeks 3–4: How LLMs Work, and Principles of Effective Prompting

Name: _____ Student No: _____ Date: _____

Instructions. This paper contains 10 multiple-choice questions. Select the **single best** answer for each. Each question is worth equal marks. Time allowed: 10 minutes. No notes or electronic devices.

1. A **token** is best described as:
 - A) Always a single word
 - B) A piece of text — a word, part of a word, a character, or punctuation
 - C) A single letter
 - D) A complete sentence
2. Why do modern systems break words into sub-word pieces rather than using a fixed list of whole words?
 - A) It makes the model smaller
 - B) It allows any word, including unseen ones, to be assembled from familiar pieces
 - C) It improves the model's grammar
 - D) It reduces the training time
3. Why does the same content typically cost more tokens in Turkish than in English?
 - A) Turkish sentences are longer
 - B) Turkish is agglutinative, and stacked suffixes fragment into several pieces
 - C) Turkish uses a different alphabet
 - D) Turkish text is processed twice
4. An assistant miscounts how many times a letter appears in a long word. The best explanation is:
 - A) The model is not good at mathematics
 - B) The model never perceived individual letters — it works on tokens
 - C) The context window was exceeded
 - D) The temperature setting was too high
5. Which statement best describes the model's core operation?
 - A) It retrieves the most relevant stored fact
 - B) It produces a probability for every possible next token, then one is chosen
 - C) It searches the internet and summarises the results
 - D) It applies a set of rules written by its developers
6. Why does the same prompt sometimes produce different answers?
 - A) The model is retrained between requests

- B) There is deliberate randomness in choosing which token comes next
 - C) The context window changes size
 - D) The model remembers the previous answer and avoids repeating it
7. Asking the same question twice and receiving a similar answer tells you:
- A) The answer is correct
 - B) The pattern is strongly represented in the training data
 - C) The model is confident
 - D) The context window is large enough
8. Which statement about hallucination is correct?
- A) It occurs when the context window overflows
 - B) It is a bug that will be removed in future versions
 - C) There is no step at which the system checks whether what it says is true
 - D) It only happens with small models
9. A colleague adds “Be accurate and only use reliable sources” to a prompt about a topic the model knows little about. What is the most likely effect?
- A) The answer becomes accurate
 - B) The model refuses to answer
 - C) The answer sounds more authoritative without becoming better informed
 - D) The context window expands
10. Output arrives with correct content but as a wall of unstructured text when you wanted three short bullets. According to the Week 4 diagnosis, which lever should you pull?
- A) Context
 - B) Task
 - C) Format
 - D) Examples